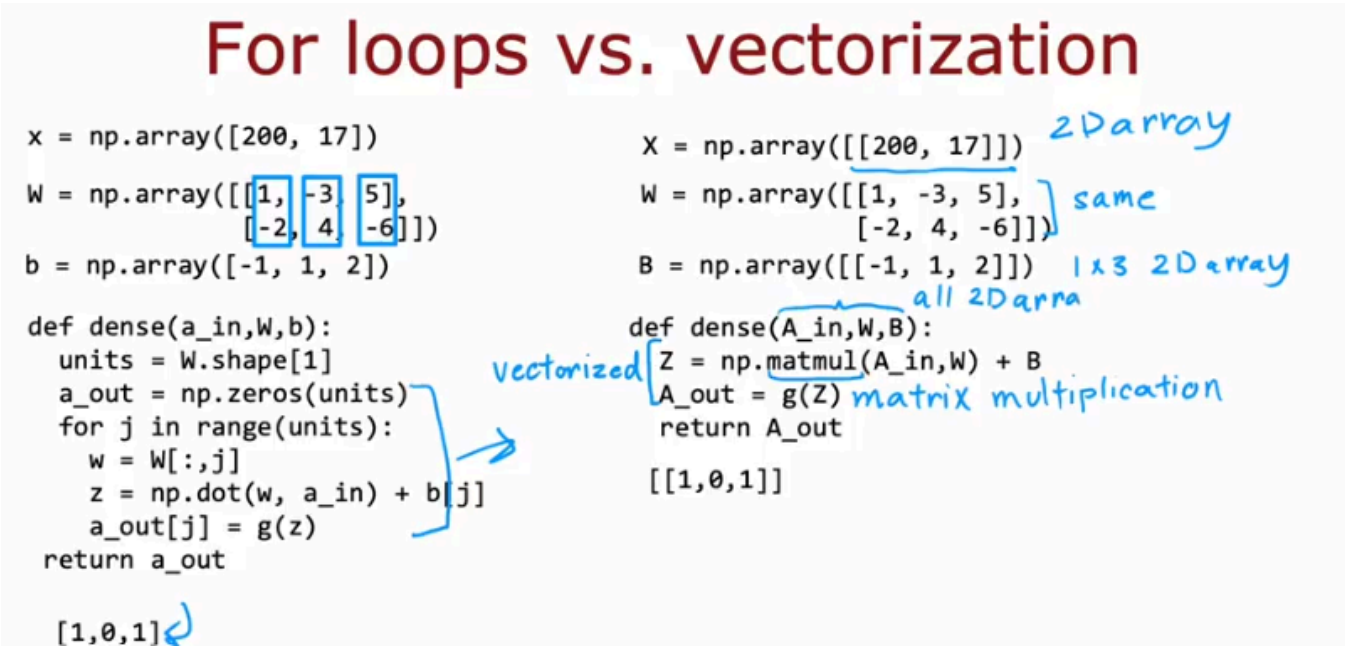
Day 15. Vectorization of neural network

Vectorization



Matrix multiplycation

> since the matrix multiplication is something very simple, we will just skip through this part and jump straight into the code

Matrix multiplication in Numpy

Matrix multiplication in NumPy

$$A = \begin{bmatrix} 1 & -1 & 0.1 \\ 2 & -2 & 0.2 \end{bmatrix} \quad A^T = \begin{bmatrix} 1 & 2 \\ -1 & -2 \\ 0.1 & 0.2 \end{bmatrix} \quad W = \begin{bmatrix} 3 & 5 & 7 & 9 \\ 4 & 6 & 8 & 0 \end{bmatrix} \quad Z = A^T W = \begin{bmatrix} 11 & 17 & 23 & 9 \\ -11 & -17 & -23 & -9 \\ 1.1 & 1.7 & 2.3 & 0.9 \end{bmatrix}$$

```
A=np.array([[1,-1,0.1],
            [2,-2,0.2]])

AT=np.array([[1,2],
            [-1,-2],
            [0.1,0.2]])

AT=A.T
```

transpose

```
W=np.array([[3,5,7,9],
            [4,6,8,0]])

Z = np.matmul(AT,W)
```

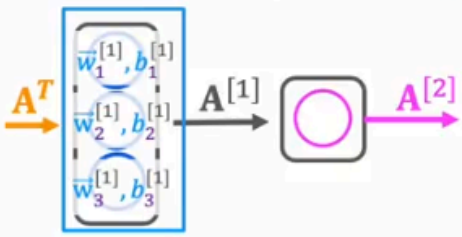
or

```
Z = AT @ W
```

result

```
[[11,17,23,9],
 [-11,-17,-23,-9],
 [1.1,1.7,2.3,0.9]
]
```

Dense layer vectorized



$$A^T = [200 \quad 17]$$

1 x 2

$$W = \begin{bmatrix} 1 & -3 & 5 \\ -2 & 4 & -6 \end{bmatrix}$$

2 x 3

$$B = [-1 \quad 1 \quad 2]$$

1 x 3

$$Z = A^T W + B$$

$$[\boxed{165} \quad \boxed{-531} \quad \boxed{900}]$$

$$z_1^{[1]} \quad z_2^{[1]} \quad z_3^{[1]}$$

$$A = g(Z)$$

$$[1 \quad 0 \quad 1]$$

A

```
AT = np.array([[200, 17]])
W = np.array([[1, -3, 5],
              [-2, 4, -6]])
b = np.array([[-1, 1, 2]])
```

a_in

```
def dense(AT,W,b):
  z = np.matmul(AT,W) + b
  a_out = g(z)
  return a_out
```

a_in

```
[[1,0,1]]
```

Optional: Numpy broadcasting tutorial

In the last example, $\mathbf{Z} = \mathbf{XW} + \mathbf{b}$ utilized NumPy broadcasting to expand the vector $\mathbf{b}$. If you are not familiar with NumPy Broadcasting, this short tutorial is provided.

$\mathbf{XW}$ is a matrix-matrix operation with dimensions $(m, j_1)(j_1, j_2)$ which results in a matrix with dimension (m, j_2). To that, we add a vector $\mathbf{b}$ with dimension $(1, j_2)$. $\mathbf{b}$ must be expanded to be a (m, j_2) matrix for this element-wise operation to make sense. This expansion is accomplished for you by NumPy broadcasting.

Broadcasting applies to element-wise operations.
Its basic operation is to 'stretch' a smaller dimension by replicating elements to match a larger dimension.

More specifically: When operating on two arrays, NumPy compares their shapes element-wise. It starts with the trailing (i.e. rightmost) dimensions and works its way left. Two dimensions are compatible when

- they are equal, or
- one of them is 1

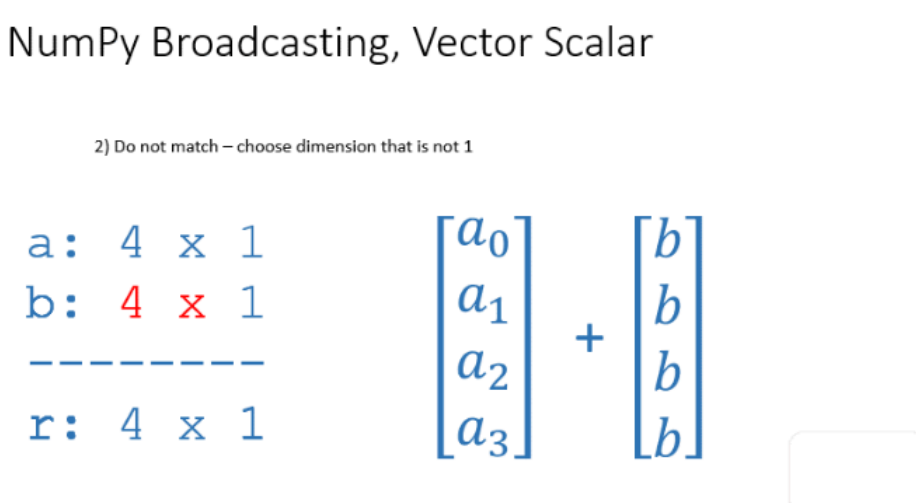
If these conditions are not met, a ValueError: operands could not be broadcast together exception is thrown, indicating that the arrays have incompatible shapes. The size of the resulting array is the size that is not 1 along each axis of the inputs.

Here are some examples:

```
a:        m x 1      1 x n      4 x 1
b:            1          1      1 x 3
-----     -----      -----      -----
result:   m x 1      1 x n      4 x 3
```

Calculating Broadcast Result shape

The graphic below describes expanding dimensions. Note the red text below:



Broadcast notionally expands arguments to match for element wise operations